

Day 2

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).

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No.	Common Hardware Issue	Typical Symptoms
1	Faulty RAM (Memory)	The computer may fail to boot, show a blank/black screen, restart repeatedly, or produce beep codes from the motherboard. Windows may also freeze or display Blue Screen errors.
2	Failed Power Supply Unit (PSU)	The PC may not turn on at all, show no power lights, or turn off suddenly. It may also restart randomly when the computer is under heavy load.
3	CPU Overheating	The computer may become very slow, freeze, restart, or shut down automatically. The CPU temperature may become unusually high, often because of a blocked heatsink, failed fan, or poor thermal paste.
4	Faulty Hard Drive/SSD	The computer may take a long time to boot, applications may freeze, and files may become corrupted or inaccessible. The system may also show disk errors or fail to start Windows.
5	Faulty Graphics Card (GPU)	The monitor may show no display, flickering, colored lines, or unusual blocks/artifacts. The computer may also crash when running games or other graphics-intensive applications.

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.

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Laptop Not Charging – Step-by-Step Troubleshooting Plan

1. Check the power outlet

Plug another device into the same electrical outlet to make sure the outlet is working. If the other device does not work, try a different outlet.

2. Inspect the charger and cable

Check the charger, cable, and connector for cuts, exposed wires, bends, burn marks, or other physical damage. If the cable or adapter looks damaged, do not use it.

3. Check the charging adapter

Feel the power adapter carefully. It should normally become slightly warm while operating. If it remains completely cold and the laptop shows no charging indication, the adapter may be faulty.

4. Check the laptop charging port

Inspect the charging port for dust, dirt, bent pins, or looseness. Make sure the charger connector fits firmly into the port.

5. Check the charging indicator

Connect the charger and look for the laptop's charging LED. If there is no charging light, check whether the adapter or charging port is making proper contact.

6. Try a known-good compatible charger

If available, test the laptop with another charger that has the **correct voltage, connector, and sufficient power rating**. If the laptop charges with the replacement charger, the original charger may be faulty.

7. Perform a power reset

Disconnect the charger and turn off the laptop. If the battery is removable, remove it. Hold the power button for about **15–20 seconds**, reconnect the battery/charger, and try turning the laptop on again.

8. Check the battery physically

If the battery is removable, inspect it for swelling, leakage, or physical damage. **Do not use a swollen or damaged battery**. If the battery is internal and the laptop still does not charge, professional servicing may be required.

9. Identify the likely fault

If the outlet, charger, cable, and charging port appear normal but the laptop still does not charge, the problem may be with the **battery, charging circuit, DC jack, or motherboard**. At this point, the laptop should be checked by a qualified technician.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).

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USB Charger Voltage Test

- 1) Set the multimeter to **DC Voltage (V—)**.
- 2) Connect the charger to a power outlet.
- 3) Connect the **black probe to GND** and **red probe to +5V (VBUS)** using a USB test/breakout adapter.
- 4) Record the voltage shown on the multimeter.
- 5) Compare it with the expected range **4.75V–5.25V**.



4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate? Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.

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ATX Power Supply 12V Rail Test – Detailed Steps

1. Turn off the PC

Shut down the computer completely and unplug the AC power cable from the wall outlet. Wait a few minutes before disconnecting the power supply cables.

2. Disconnect the ATX connector

Open the PC case and disconnect the main 24-pin ATX power connector from the motherboard. The connector should be disconnected before testing the individual wires.

3. Set the multimeter

Turn the multimeter on and select DC Voltage (V_{DC}). If the meter uses manual ranges, select the 20V DC range.

4. Identify the wires

On a standard ATX power connector:

- Yellow wire = +12V
- Black wire = Ground (GND)

The yellow wire carries the 12V supply, while the black wire provides the ground reference.

5. Connect the probes

Place the black probe on a black GND wire and the red probe on the yellow +12V wire.

For a real PSU test, the PSU must be powered appropriately while the connector is disconnected from the motherboard. Do not allow the two probes to touch each other or short adjacent pins.

6. Read the multimeter

Check the voltage displayed on the multimeter.

For example:

Measured voltage = 12.08V

This is close to the expected 12V output.

7. Compare the reading

For this assignment, use 11.5V as the warning threshold.

- 12.08V → Normal
- 11.5V or higher → Generally acceptable for this assignment
- Below 11.5V → Possible PSU problem

A low 12V reading can indicate that the power supply is weak or faulty and may be unable to provide stable power to components such as the CPU and graphics card.

8. Relate it to the random shutdown

If the PC is randomly shutting down and the 12V rail measures significantly below 11.5V, the PSU should be suspected. An unstable or insufficient power supply can cause sudden shutdowns, restarts, or system instability.

